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AI as a Thinking Partner

Why the dominant approach produces faster workers with weaker judgment — and what the evidence says about the alternative

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EXECUTIVE SUMMARY

Use AI to surface, challenge, and validate your own reasoning — and treat your judgment as the authoritative conclusion.

Enterprises spend an estimated $30–40 billion annually on AI tools; 95 percent report no measurable P&L impact. The uncomfortable explanation is that organizations taught people to go faster without teaching them to think better — and the trial evidence shows a different design produces a different outcome.

01 Interaction design decides the outcome: in randomized trials, standard AI use cut unassisted performance by 17 percent while a reasoning-first design doubled learning gains.

02 Two tracks, two timelines: installing the pre-mortem habit is a 60–90 day behavioral change; recovering degraded reasoning capacity is a 12–36 month effort. Conflating them breaks the program.

03 Three layers must activate together — training design, usage configuration, and institutional accountability — because individual adoption without reinforcement is structurally penalized.

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The $30 billion bet that is backfiring

Enterprises have spent an estimated $30–40 billion annually deploying AI tools. Ninety-five percent report no measurable P&L impact. The dominant explanation is poor workflow integration or unclear ROI. The actual explanation is simpler and more uncomfortable: organizations taught people how to go faster without teaching them how to think better — and in doing so, began quietly eroding the judgment that justified the investment.

The cause is training design, and the consequence is measurable. Two peer-reviewed RCTs now bracket the outcome space precisely. Students using standard AI with no guidance scored 17 percent worse on independent assessments than a control group that used no AI at all. Students using AI designed to require reasoning first scored more than twice as well as a control group using best-practice human instruction. Same tool, different interaction design. That gap separates what most organizations are producing today from what is demonstrably achievable.

The problem begins before the workforce. By the time an employee arrives with entrenched AI shortcuts, those habits were formed in school — under prohibition, without guidance, optimized for the only visible goal: get the task done. Seventy-eight percent of knowledge workers bring personal, unsanctioned AI tools to work, and roughly 70 percent of active AI users have received no training from their employer. The habit vacuum fills with whatever people teach themselves under pressure — and the research now shows exactly what that produces.

The design gap that defines the outcome

Bastani et al. (PNAS, 2025) randomized approximately 1,000 students across three conditions: standard AI, a guardrailed tutor, or no AI. Standard AI users scored 17 percent worse on unassisted assessments than the control group. The guardrailed tutor — the same underlying model, redesigned to require students to reason through hints before receiving answers — eliminated the deficit entirely. The harm came from a specific interaction design that let users bypass their own reasoning, not from the technology itself.

EXHIBIT 1

Interaction design decides the outcome

–17% / +2×

The measured range between ungoverned AI use (17 percent worse on unassisted assessments) and a reasoning-first design (twice the learning gains of best-practice active learning) in peer-reviewed randomized trials.

Bastani et al., PNAS 2025; Kestin et al., Scientific Reports 2025 (see Notes).

Kestin et al. (Scientific Reports, 2025) randomized 194 undergraduates to a custom AI tutor or best-practice active learning: peer instruction, small groups, real-time instructor feedback. The AI tutor produced more than twice the learning gains in less time — 49 minutes versus 60 — with statistical significance of $p < 10^{-8}$. The comparison was against active learning, the current gold standard of human instruction, not a passive lecture.

That gap is a design decision. Most institutions are making it by default, in the wrong direction.

Scope limitation: The Kestin study used Harvard undergraduates studying introductory physics over two weeks. The researchers explicitly state results target middle-order cognitive skills and cannot be presumed to extend to higher-order synthesis or professional judgment. The effect is real and robust — and conditional on the deliberate interaction design this paper argues for.

The real state of AI use

Condition	Share	Source
U.S. students who consider AI essential for success	**65%**	HEPI 2025
UK university students using AI for assessments	**88%**	HEPI 2025
Teachers who allow any AI use	**38%**	RAND 2025
Teachers who received training on AI misuse response	**28%**	Dwyer & Laird 2024
U.S. knowledge workers using AI at work	**45%**	Gallup Q3 2025
Bringing personal, unsanctioned AI tools	**78% of AI users**	Microsoft / LinkedIn WTI 2024
Received any AI training from employer	**30% of AI users**	Microsoft / LinkedIn WTI 2024
Would not stop using AI even if banned	**~48%**	SANS, Oct 2024
Share confidential data with AI tools without approval	**38%**	CybSafe / NCSA 2024

If 45 percent of the U.S. workforce uses AI at work and only 30 percent received any employer training, approximately 70 percent of all active AI users are operating on a self-taught framework. That framework was reverse-engineered from the tool's interface, from peer behavior, and from the pressure to produce output quickly and invisibly. The top reported use cases — consolidating information (42 percent), generating ideas (41 percent) — are output-first behaviors. Neither requires interrogating assumptions or sustaining the reasoning effort that durable expertise demands.

What the research shows

Study	Population	Method	Key Finding	Evidence Quality
Bastani et al. (2025), PNAS	~1,000 students	RCT	Standard AI: −17% on exams. Guardrailed tutor: parity restored.	High — peer-reviewed RCT, pre-registered
Kestin et al. (2025), Scientific Reports	194 undergraduates	RCT	Reasoning-first AI tutor: 2× learning gains vs. best-practice active learning, $p < 10^{-8}$	High — peer-reviewed RCT; scope limited to introductory physics
Liu et al. (2025), Scientific Reports	3,562 workers, 4 studies	Experimental	AI raised output quality but reduced intrinsic motivation 11% and increased boredom 20% on solo tasks	High — peer-reviewed, four replications
Lee et al. (2025), Microsoft / CMU	319 knowledge workers	Survey	Higher AI confidence → less critical thinking applied; higher domain confidence → more	Moderate — peer-reviewed CHI; self-report
Gerlich (2025), MDPI Societies	666 participants	Survey + interviews	Frequent AI use negatively correlated with critical thinking; balanced use attenuated the effect	Moderate — peer-reviewed; no objective performance measure
Kosmyna et al. (2025), MIT Media Lab	54 adults (18 in session 4)	EEG preprint	LLM group: up to 55% reduced neural connectivity; 83% memory recall deficit	Low — preprint; underpowered; formal methodological concerns

Note on the MIT preprint: Formal peer commentary identified five methodological problems including underpowered sample (n=18 for the key session 4 finding), absent pre-registration, and EEG inconsistencies. The lead author explicitly asks the work not be characterized as evidence of permanent harm. It is cited as directionally consistent only and plays no role in the recommendation architecture below.

What Gerlich's nuance means operationally

Gerlich's finding that balanced use attenuates the negative effect describes what the accountability partner model produces in practice. Participants who used AI for administrative tasks while preserving reasoning effort for complex and evaluative work did not show the critical-thinking decline seen in heavy users. That is a specific, buildable behavioral pattern.

Why professionals degrade on complex tasks

Dell'Acqua et al. (Research Policy, 2024) ran a pre-registered field experiment with 758 BCG consultants.[5] For within-frontier tasks, AI users completed 12.2 percent more tasks, worked 25 percent faster, and produced outputs rated more than 40 percent higher in quality. For one task selected outside the frontier, they were 19 percentage points less likely to reach a correct solution than those without AI. Degradation was concentrated in consultants who adopted AI outputs without interrogating them — *blind adoption* in the researchers' language. The tool gave no signal it was operating beyond its capability. Institutional note: BCG co-designed a study finding BCG consultants perform better with AI. Pre-registration and multi-institutional authorship partially mitigate that interest.

Tool-level overconfidence

The calibration problem is not only behavioral. Studies across GPT-4, GPT-3.5, LLaMA2, and PaLM 2 find systematically high calibration error — high-confidence outputs regardless of accuracy. A controlled study of 257 medical students making 3,855 diagnostic decisions found a transparency paradox: AI explanations improved decisions when correct (+6.3 points) but worsened them when wrong (−4.9 points), because AI generates equally persuasive explanations regardless of recommendation quality.[6] Domain expertise is the prerequisite for that interrogation — the floor beneath it, not a substitute for it.

The evidence gaps this paper cannot close

No study directly compares cognitive outcomes between formally trained workers and self-taught BYOAI users. No enterprise-scale RCT has tested reasoning-first versus answer-first AI interaction for professional knowledge workers. Both are real gaps. The recommendations below are high-confidence inference — not established proof. Closing these gaps is the applied research agenda the field needs.

Why the usual responses fail

Prohibition does not prevent AI use. It prevents guided use. The Bastani and Kestin studies were not comparing AI against no AI — they were comparing ungoverned AI against carefully designed AI. Prohibition produces the ungoverned condition at scale, then exports it to the workforce.

The deeper cost is the force multiplier surrendered. Kestin's doubling of learning gains came against best-practice instruction — a strong control — so banning AI, or allowing undesigned access, forfeits a measured benefit, not a hypothetical one.

The security cost is separate and compounding. Shadow AI breaches cost $670,000 more per incident than standard breaches and account for 20 percent of all AI-related breaches. The average enterprise sees 223 incidents per month of sensitive data submitted to unvetted AI tools, a figure that doubled year over year. Ninety percent of security leaders use unapproved tools themselves. Prohibition drives this exposure underground rather than reducing it. Underneath both the ban and the ungoverned rollout, the same structural failures recur:

- **Completion is the only frame.** No institution has provided a frame that makes interrogating AI's output the point. In school the goal is submitting the assignment. At work the goal is closing the task. The tool's interface reinforces both.
- **No baseline for reasoning quality.** Activity data is captured; reasoning quality is not. The harm accumulates in a measurement blind spot. Organizations cannot address what they are not measuring.

- **No designed friction.** The Bastani finding is precise: the harm came from an interaction design that let users bypass reasoning. The fix is designed friction. Prohibition removes access without replacing it with productive struggle.
- **The tool is trusted where it should be interrogated.** The transparency paradox means AI generates equally persuasive explanations for correct and incorrect outputs. Training must compensate for what the tool structurally cannot signal.
- **No accountability loop.** A worker who accepts AI output without interrogation and one who rigorously pressure-tests it look identical on any activity dashboard. Without an external signal that reasoning quality matters, throughput is the only visible standard — and the collective action dynamic penalizes individual departure from it.

Why the stakes are higher than they appear

The wage premium

Autor and Thompson's 2025 NBER working paper provides the labor economics frame: when automation eliminates non-expert tasks from an occupation, the wage premium for the remaining expert tasks rises; when it eliminates expert tasks, wages fall.[22] AI is automating answer-generation, routine synthesis, and information consolidation — the non-expert component of knowledge work — so the scarcity value of judgment, evaluation, and reasoning rises accordingly. Workers who preserve those capabilities through the accountability partner model capture that premium; workers who offload judgment entirely give up the tasks whose value is rising. Because that benefit is personal and wage-linked, it is easier to sustain than a distant, collective payoff.

The agentic AI horizon

Generative AI still involves humans in both task execution and judgment. Agentic AI — autonomous systems that plan and complete multi-step workflows with minimal human input — shifts the human role almost entirely to goal-setting, output evaluation, and exception handling. That is precisely the task stewardship capacity that deteriorates fastest under unguided AI use. Workers who trained themselves to bypass reasoning under generative AI will be underprepared for the one distinctly human role in the architecture that is already being built. The training decisions being made right now are more consequential than they currently appear.

A better standard: AI as accountability partner

Working standard — defined for this paper:

AI is functioning as a cognitive partner when it is used to surface, challenge, and validate a user’s own reasoning — not replace it — and when the user’s judgment, informed by AI interrogation, is treated as the authoritative conclusion.

Treat every AI output as a starting position for thinking. The default interaction is interrogative: pressure-test the plan before producing it; identify the weakest assumption before making the argument; raise the counterarguments before writing the brief.

When does this standard apply?

Task Type	Consequence Level	Recommended Approach
Within AI's reliable frontier	Low — routine, well-defined	Full delegation appropriate
Within AI's reliable frontier	High — strategic, consequential	Verify outputs; check reasoning chain
Outside or near AI's frontier	Any	Pre-mortem protocol required before AI engagement
Unknown frontier position	High	Treat as outside-frontier; pre-mortem required

The pre-mortem prompt protocol

Before engaging AI on any high-judgment task, the worker completes three steps without AI: (1) state the hypothesis or recommendation they expect to reach and why; (2) identify the two or three assumptions that most need to be true; (3) articulate what evidence would change their view. AI is then introduced to interrogate the pre-mortem — challenge the assumptions, raise counterarguments, locate the weakest reasoning. The worker's judgment, challenged by AI, is the conclusion.

This has independent validation. Ma et al. (CHI 2024) developed a structurally identical intervention — 'Thinking the Opposite' — and found it improved self-confidence calibration and directly reduced AI over-reliance. Klein's original pre-mortem technique has its own validated lineage in team decision-making.

Two qualifications: First, AI-generated challenges carry the same confident tone regardless of correctness. The protocol improves the user's reasoning posture — it does not resolve the tool's calibration problem. For tasks outside the user's domain expertise, AI challenges must be validated against external domain knowledge, not accepted as authoritative. Second, self-directed habit formation without external reinforcement has low sustained adoption under time pressure. This protocol changes the default when embedded in institutional processes. Individual adoption without that embedding is fragile.

Two tracks, not one timeline

Installing the pre-mortem habit is a 60–90 day behavioral change. Recovering degraded reasoning capacity is a 12–36 month restorative effort. Conflating the two produces a program that overpromises on the slower problem and underserves the faster one.

EXHIBIT 3

Two tracks, two problems, two timelines

Track 1 — Protective (60–90 days)
Workers with intact reasoning capacity who have not yet formed deep shortcut habits: install the pre-mortem practice.

Track 2 — Restorative (12–36 months)
Recovering degraded reasoning capacity: a slower restorative effort that must not be promised on the fast track's timeline.

The audit is a capability triage. Establishing which workers are in Track 1 versus Track 2 requires two components: a standard tool use inventory, and domain-relevant judgment tasks performed without AI assistance and evaluated for reasoning quality. Without the second component, training is deployed without knowing whether it is protecting or restoring — two different problems requiring different interventions at different timescales. The most urgent population is your most senior judgment-dependent roles that have been heaviest BYOAI users. They represent the highest organizational risk and the longest restoration timeline.

Track 2 restorative curriculum at professional scale does not yet exist. This is a named research gap. Organizations running Track 2 programs are doing so without validated playbooks. The research agenda that would close this gap — enterprise-scale trials of structured AI-removal practice with expert judgment feedback — is the work the field needs to prioritize.

The three-layer change model

All three layers must activate simultaneously. The collective-action problem means individual adoption ahead of institutional reinforcement is structurally penalized, and the temporal-discount dynamic means workers take the shortcut under pressure without external enforcement.

EXHIBIT 2

Three layers must activate simultaneously

1. **Training design — CHRO and L&D**
Frame it as a decision-quality initiative, run capability triage first, introduce the pre-mortem as a performance advantage, teach the tool's limits, and evaluate reasoning rather than output.

2. **Usage configuration — IT Security and Operations**
Governance before control: provision rather than ban, make the pre-mortem an organizational norm for high-judgment tasks, and audit full-delegation patterns.

3. **Institutional accountability — managers and executives**
Make reasoning quality visible in review, track task-stewardship capacity, reward reasoning in performance criteria, and protect the pilot from the existing measurement system.

Layer 1 — Training design

Owner: *CHRO and L&D function (Schools: curriculum coordinators and teacher professional development leads)*

- **Frame this as a decision quality initiative, not an AI initiative.** 'AI initiative' triggers compliance instincts and turf conflict. 'Decision quality initiative' is a language every P&L owner speaks. The outcome being pursued is better judgment — AI is the context, not the product.
- **Run capability triage before training.** Establish what employees are using and for what tasks — with disclosure decoupled from discipline. Augment tool inventory with domain judgment tasks performed without AI to identify Track 1 versus Track 2 populations. Deploying the same training to both wastes resources and misses the restorative population entirely.
- **Introduce the pre-mortem protocol as a performance advantage, not a mandate.** The Dell'Acqua finding is the self-interest argument: consultants who interrogated AI outputs outperformed blind adopters with identical credentials. That is a performance case, not a compliance case.
- **Teach the tool's limits explicitly.** LLMs are structurally overconfident. Training must address what AI cannot reliably signal: the boundaries of its own competence. Teach workers to treat AI confidence level as a non-diagnostic signal and to apply domain expertise as the primary validity check on AI-generated challenges.
- **Evaluate reasoning, not output.** Assessment criteria should center on how well the learner articulated the problem, how rigorously they interrogated AI responses, and how defensible their final judgment is. Measuring only output completion reinforces the throughput frame the training is designed to displace.

Layer 2 — Usage pattern configuration

Owner: *IT Security and Operations, in coordination with CHRO (Schools: IT and academic integrity office)*

- **Governance before control.** Banning BYOAI produces the worst outcome: continued use with added concealment. Provision reduces the incentive for shadow behavior by meeting the underlying need through sanctioned alternatives. The BYOD parallel is instructive: formal policies reduce but do not eliminate unauthorized use. No AI-specific evidence of equivalent design currently exists; the BYOD analog is a structural parallel, not direct proof.
- **Implement the pre-mortem protocol as an organizational norm for high-judgment tasks.** Tasks involving strategy, significant resource decisions, client-facing analysis, or risk assessment require cognitive priming before AI engagement. It sets the sequence for high-judgment work.
- **Audit full-delegation patterns.** Identify tasks handed entirely to AI without a verification or reasoning loop. For tasks involving judgment or ambiguous tradeoffs, full delegation without a reasoning layer is risk accumulation — as the Dell'Acqua outside-frontier findings demonstrate directly.

Layer 3 — Institutional accountability

Owner: *Direct managers and executive leadership (Schools: department heads and academic leadership)*

This layer is not optional. Without it, individual adoption is penalized and the training investment decays under throughput pressure within 60–90 days.

- **Make reasoning quality visible.** Review processes for AI-assisted work must probe the reasoning behind conclusions, not only the conclusions. The operative question is not 'did AI help produce this?' It is 'what is the reasoning behind this conclusion, and how was it validated?' The distinction between conclusions reflecting the employee's judgment and those reflecting AI output with an approval signature must become visible to management.
- **Track task stewardship capacity.** Define this as the ability to set appropriate goals for AI, evaluate its outputs, and make sound independent judgments when AI is inadequate. Assess it before and after training interventions. It is a leading indicator — reasoning atrophy shows up in decision quality only after it is already significant.
- **Signal that reasoning quality is rewarded.** Explicit recognition of reasoning quality in performance reviews, promotion criteria, and project retrospectives changes the present-moment calculus. This is the mechanism that makes the accountability partner model individually rational, not just institutionally desirable.
- **Protect the pilot from the existing measurement system.** Teams practicing the pre-mortem protocol will produce work more slowly than peers who are not — in the short term, against any throughput metric. In the first 90 days, the executive sponsor's job is to hold the measurement question open long enough for a quality signal to emerge. That is the minimum viable executive support the initiative requires.

Call to action: implementation without a mandate

The three-layer model works once an organization has decided to act, and most have not. An implementation model that requires full organizational backing stalls before it starts.

Durable change at this scale moves by demonstration. The practical sequence:

Phase	Timeline	Action
Name the owner	Week 1	Assign a single executive — CHRO with CIO as co-sponsor — explicitly accountable for all three layers. Not a working group. One person who owns the outcome.
Run capability triage	Month 1	Grant amnesty for BYOAI disclosure. Inventory tool use. Assess reasoning depth through AI-removed judgment tasks. Identify Track 1 versus Track 2 populations. This is the only way to know what you are actually dealing with.
Pilot one workflow	Months 2–3	Select one function where reasoning quality is already consequential — strategy, risk, client advisory, product decisions. Run the pre-mortem protocol for one quarter. Define decision quality criteria now, before the pilot begins.
Rewrite one rubric	Months 3–6	Add task stewardship as an assessed competency in one function. Make reasoning quality visible in one review cycle. This is the signal that the institutional incentive has changed.
Evaluate and scale	Month 6	If the pilot shows signal, you have the internal proof of concept that makes scaling tractable. If it does not, you have spent 90 days and learned something real before committing the enterprise.
Culture shift	18–36 months	From first pilot to embedded norm in the functions that matter most. This is what durable culture change costs when it has to prove its value rather than assume it. It is an accurate forecast, not a lack of ambition.

The one thing that kills it: treating Layer 3 as downstream of Layers 1 and 2. If reasoning quality is not made visible and rewarded at the same time as the training rollout, the protocol decays within 60–90 days and the organization returns to its prior baseline. Put Layer 3 in place from the start, as the foundation.

What the choice comes down to

The −17 percent and the +2× are the same technology under two interaction designs. Defaulting to the first accepts a specific, measured cost while a specific, measured benefit sits available for the same tools. Closing that gap does not require new technology; it requires a different definition of what AI is for — a starting position for thinking rather than a finished answer.

That definition is not new. The professional norm it depends on — you own your conclusions, and judgment stays visible and valued — predated AI by decades. AI eroded it in a measurement blind spot, faster than most organizations noticed. So the work here is recovery: it asks people to act consistently with a standard they already hold. The research now shows what each choice produces, and the choice is institutional.

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